

## Advanced Data Science - Homework 2

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**Exercise 1.** Let  $\tau \in [0, 1]$ . Prove that for any  $z_1, \dots, z_m, z \in \mathbb{R}$ ,

$$z \leq \hat{Q}_{m+1}^\tau(z_1, \dots, z_m, z) \iff z \leq \hat{Q}_m^{\tau(1+1/m)}(z_1, \dots, z_m).$$

*Solution.* Suppose that  $z \leq \hat{Q}_{m+1}^\tau(z_1, \dots, z_m, z)$ . We have

$$\begin{aligned} \hat{Q}_{m+1}^\tau(z_1, \dots, z_m, z) &= \inf \left\{ t \in \mathbb{R} : \frac{1}{m+1} \left( \sum_{i=1}^m \mathbb{1}_{\{z_i \leq t\}} + \mathbb{1}_{\{z \leq t\}} \right) \geq \tau \right\} \\ &= \inf \left\{ t \in \mathbb{R} : \sum_{i=1}^m \mathbb{1}_{\{z_i \leq t\}} + \mathbb{1}_{\{z \leq t\}} \geq (m+1)\tau \right\} \\ &= \inf \underbrace{\left\{ t \in \mathbb{R} : \sum_{i=1}^m \mathbb{1}_{\{z_i \leq t\}} + \mathbb{1}_{\{z \leq t\}} \geq \lceil (m+1)\tau \rceil \right\}}_Q. \end{aligned}$$

The last equality holds because  $\sum_{i=1}^m \mathbb{1}_{\{z_i \leq t\}} + \mathbb{1}_{\{z \leq t\}}$  is an integer. Similarly,

$$\begin{aligned} \hat{Q}_m^{\tau(1+1/m)}(z_1, \dots, z_m) &= \inf \left\{ t \in \mathbb{R} : \frac{1}{m} \sum_{i=1}^m \mathbb{1}_{\{z_i \leq t\}} \geq \tau(1+1/m) \right\} \\ &= \inf \underbrace{\left\{ t \in \mathbb{R} : \sum_{i=1}^m \mathbb{1}_{\{z_i \leq t\}} \geq \lceil (m+1)\tau \rceil \right\}}_{Q'}. \end{aligned}$$

If  $\sum_{i=1}^m \mathbb{1}_{\{z_i \leq t\}} \geq \lceil (m+1)\tau \rceil$ , then  $\sum_{i=1}^m \mathbb{1}_{\{z_i \leq t\}} + \mathbb{1}_{\{z \leq t\}} \geq \lceil (m+1)\tau \rceil$ . Hence,  $Q' \subset Q$ , or

$$\hat{Q}_{m+1}^\tau(z_1, \dots, z_m, z) \leq \hat{Q}_m^{\tau(1+1/m)}(z_1, \dots, z_m).$$

That means, if  $z \leq \hat{Q}_{m+1}^\tau(z_1, \dots, z_m, z)$ , then  $z \leq \hat{Q}_m^{\tau(1+1/m)}(z_1, \dots, z_m)$ .

We will prove the converse by contrapositive. Suppose that  $z > \hat{Q}_{m+1}^\tau(z_1, \dots, z_m, z)$ . By monotonicity of the function  $t \mapsto \sum_{i=1}^m \mathbb{1}_{\{z_i \leq t\}} + \mathbb{1}_{\{z \leq t\}}$ , then there exists  $t^* < z$  such that

$$\sum_{i=1}^m \mathbb{1}_{\{z_i \leq t^*\}} + \mathbb{1}_{\{z \leq t^*\}} \geq \lceil (m+1)\tau \rceil.$$

But  $\mathbb{1}_{\{z \leq t^*\}} = 0$ . Hence,

$$\sum_{i=1}^m \mathbb{1}_{\{z_i \leq t^*\}} \geq \lceil (m+1)\tau \rceil.$$

Therefore,  $t^* \in Q'$  or  $z > t^* \geq \hat{Q}_m^{\tau(1+1/m)}(z_1, \dots, z_m)$ .

**Exercise 2.** Let  $\tau \in [0, 1]$ . Prove that if  $Z_1, \dots, Z_m$  are exchangeable, then

$$\tau \leq \mathbb{P} \left( Z_i \leq \hat{Q}_m^\tau(Z_1, \dots, Z_m) \right)_{Z_i \text{ continuous}} \leq \tau + \frac{1}{m}.$$

*Solution.* Using a similar argument as in Exercise 1,

$$\hat{Q}_m^\tau(Z_1, \dots, Z_m) = \inf \left\{ t \in \mathbb{R} : \sum_{j=1}^m \mathbf{1}_{\{Z_j \leq t\}} \geq \lceil m\tau \rceil \right\}$$

By exchangeability, for any Borel set  $S \subset \mathbb{R}^m$  and any permutation  $\sigma$  of  $\{1, \dots, m\}$ ,

$$\mathbb{P}((Z_1, \dots, Z_m) \in S) = \mathbb{P}((Z_{\sigma(1)}, \dots, Z_{\sigma(m)}) \in S).$$

Choose  $S = \{(z_1, \dots, z_m) \in \mathbb{R}^m : z_1 \leq \hat{Q}_m^\tau(z_1, \dots, z_m)\}$ . We have,

$$\begin{aligned} \mathbb{P} \left( Z_1 \leq \hat{Q}_m^\tau(Z_1, \dots, Z_m) \right) &= \mathbb{P}((Z_1, \dots, Z_m) \in S) \\ &= \mathbb{P}((Z_2, Z_1, Z_3, \dots, Z_m) \in S) \\ &= \mathbb{P} \left( \{(Z_2, Z_1, Z_3, \dots, Z_m) : Z_2 \leq \hat{Q}_m^\tau(Z_2, Z_1, Z_3, \dots, Z_m)\} \right) \\ &= \mathbb{P} \left( Z_2 \leq \hat{Q}_m^\tau(Z_1, \dots, Z_m) \right) \end{aligned}$$

The last equality holds because  $\hat{Q}_m^\tau$  is symmetric in its arguments. Continuing the process, we get the fact that  $\mathbb{P} \left( Z_i \leq \hat{Q}_m^\tau \right)$  is the same for every  $i \in \{1, \dots, m\}$ . By definition of  $\hat{Q}_m^\tau := \hat{Q}_m^\tau(Z_1, \dots, Z_m)$ , we have

$$\sum_{j=1}^m \mathbf{1}_{\{Z_j \leq \hat{Q}_m^\tau\}} \geq \lceil m\tau \rceil.$$

Thus we have the lower bound

$$\begin{aligned} \mathbb{P} \left( Z_i \leq \hat{Q}_m^\tau \right) &= \frac{1}{m} \sum_{j=1}^m \mathbb{P} \left( Z_j \leq \hat{Q}_m^\tau \right) \\ &= \frac{1}{m} \mathbb{E} \left[ \sum_{j=1}^m \mathbf{1}_{\{Z_j \leq \hat{Q}_m^\tau\}} \right] \\ &\geq \frac{\lceil m\tau \rceil}{m} \geq \frac{m\tau}{m} = \tau. \end{aligned}$$

Now we prove the upper bound. Using exchangeability again for  $S = A \times \mathbb{R}^{m-1}$  for some Borel set  $A \subset \mathbb{R}$ , we have

$$\mathbb{P}(Z_1 \in A, Z_2 \in \mathbb{R}, \dots, Z_m \in \mathbb{R}) = \mathbb{P}(Z_2 \in A, Z_1 \in \mathbb{R}, Z_3 \in \mathbb{R}, \dots, Z_m \in \mathbb{R}).$$

Hence,  $\mathbb{P}(Z_1 \in A) = \mathbb{P}(Z_2 \in A)$ . Continuing the process, we get the fact that  $Z_1, \dots, Z_m$  are identically distributed. Given that  $Z_i$  is continuous, all  $Z_1, \dots, Z_m$  are continuous. Hence,  $Z_i - Z_j$  is continuous, then  $\mathbb{P}(Z_i = Z_j) = 0$  for all  $i \neq j$  i.e.  $X_1, \dots, X_m$  are all distinct with probability 1. Therefore,

$$\sum_{j=1}^m \mathbf{1}_{\{Z_j \leq \hat{Q}_m^\tau\}} = \lceil m\tau \rceil$$

with probability 1. Thus,

$$\mathbb{P} \left( Z_i \leq \hat{Q}_m^\tau \right) = \frac{\lceil m\tau \rceil}{m} \leq \frac{m\tau + 1}{m} = \tau + \frac{1}{m}.$$